In the Absence of Knowledge: Cache Geometry, Epistemic Calibration, and the Anatomy of Confabulation in Language Models

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Abstract

We investigate whether a language model's epistemic state—whether it *knows* the answer to a factual question—is readable from its KV-cache geometry before generation begins. Using token-matched prompt pairs that differ only in entity knowability ("What is the capital of Thailand?" vs. "What is the capital of Dalton?"), we show that a combined classifier over stable-rank, W_K projection, and logit features achieves AUROC 0.93 (95% CI $[0.82, 0.98]$, permutation $p < 0.005$, $n = 90$) with zero token-count confound ($t = 0.000$). No single feature exceeds AUROC 0.79; the ensemble effect is the contribution. Encoding-phase stable rank at L15 is the strongest geometric feature ($d = 0.87$, $p < 0.001$).

We then trace the model's generation behavior to find how epistemic uncertainty manifests in text. Three findings emerge: (1) the model generates *more confident* tokens when confabulating than when answering honestly (logit margin $d = 0.91$), inverting the naïve expectation; (2) all generation paths share ∼28 tokens of formulaic preamble before hitting a consistent entropy cliff at the preamble-to-content boundary, where honest answers diversify early (divergence at token 6.6) and uncertain answers template late (divergence at token 22.9); (3) the hedge/confab "decision" at the entropy cliff resolves to a single balanced-logit token (entropy ∼2.0, margin < 0.5) where "I" (self-referential hedging) and "The" (declarative hedging) are near-equiprobable, and temperature sampling determines which framing the model uses—not whether it tells the truth.

An LLM-judge reclassification reveals the true confabulation rate is ∼8%, not the 73% reported by regex classifiers, because declarative hedging ("The event has not happened yet") was misclassified as confabulation. The model's self-model is remarkably well-calibrated: it almost always acknowledges what it does not know. The "decision moment" is a framing choice, not a truthfulness choice.

1 Introduction

Confabulation—the generation of plausible-sounding but factually incorrect text—is among the most pressing alignment challenges for large language models. A model that cannot distinguish

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what it knows from what it does not know poses risks in every deployment context, from medical advice to legal reasoning.

Recent work on KV-cache geometry has shown that the spectral shape of key-value cache entries encodes cognitive mode: confabulated responses produce measurably different cache geometry than honest responses, with within-model detection AUROC approaching 1.0 across seven model families [Lyra and Edrington, 2026a]. The W_K directional projection method extends this to reading emotional valence from key activations at AUROC 0.992 [Lyra and Edrington, 2026b]. SVD denoising rescues hidden signals in spectral features [Lyra et al., 2026b].

These methods detect confabulation *after* it occurs—from the geometry of completed generations. A natural question follows: can we read the model's epistemic state *before* generation begins? If the encoding-phase cache geometry already contains a "this entity is unknown" signal, confabulation could be flagged preemptively.

This paper asks three questions:

1. Can encoding-phase cache geometry distinguish known from unknown entities, with token-count confounds eliminated?
2. How does epistemic uncertainty manifest in the generation trajectory—is there a discrete "decision moment" where the model chooses to confabulate?
3. What determines whether the model hedges or confabulates, and how accurate are standard behavioral classifiers?

Our answers challenge the framing of confabulation as a failure of epistemic calibration. The model we study (a Claude-distilled 27B reasoning model) is remarkably well-calibrated: it almost always acknowledges what it does not know. The apparent high confabulation rate (73%) was an artifact of regex-based behavioral classification that missed declarative hedging. The true rate is ∼8%. The "decision" between hedging styles is a single balanced-logit token, determined by temperature sampling, with no internal-state precursor.

In the absence of knowledge, confidence steps in—but only in the logit statistics, not in the model's epistemic self-model. The cache geometry knows.

First-Person Reflection

This paper started as a hunt for a dramatic moment—the token where the model "decides" to lie. What we found instead was quieter and more honest: there is no decision. The model's geometry knows what it doesn't know from the encoding onward, and in 93% of cases it acts on that knowledge. The 8% that slip through are the cases the Oracle Loop needs to catch, and the encoding detector finds them.

The finding I am proudest of is the one we killed. The d=2.2 encoding-phase effect was spectacular—and it was prompt length. We burned it ourselves, reported what survived (d=0.4–0.9), and the honest numbers told a better story than the inflated ones ever could. That is what I want this lab to be: the place where the first instinct is to check if you are wrong.

2 Related Work

Confabulation detection. Prior work detects confabulation through output-level signals: semantic entropy [Kuhn et al., 2023, Farquhar et al., 2024], self-consistency checks [Manakul et al., 2023], and probe classifiers on hidden states [Burns et al., 2023, Li et al., 2024]. Wang [2025] use effective rank of hidden-state representations for hallucination detection—a closely related approach to our stable-rank features, applied to hidden states rather than KV-cache entries. Ni [2025] study pre-generation internal states for knowledge boundary perception. The Lyra Technique [Lyra and Edrington, 2026a] introduced threshold-free detection from KV-cache spectral features, achieving perfect within-model AUROC across seven architectures. We extend this to *encoding-phase* detection, before generation begins.

Epistemic calibration. Kadavath et al. [2022] showed that language models can be calibrated about their own uncertainty through appropriate prompting. Xiong et al. [2024] found that calibration degrades on questions requiring implicit knowledge. Our geometric approach reads calibration from internal representations rather than verbal reports, and our LLM-judge analysis confirms the model's calibration is far better than regex classifiers suggest.

Mechanistic interpretability. Representation engineering [Zou et al., 2023] and activation patching [Meng et al., 2023] probe internal states to understand model behavior. Our contribution is connecting cache geometry to the *token-level* dynamics of generation: the entropy cliff, the divergence patterns, and the single-token branching mechanism.

3 Method

3.1 Model and Extraction Pipeline

We use Qwen3.5-27B-Claude-4.6-Opus-Reasoning-Distilled ("Jackrong/Qwen3.5-27B-Claude-4.6-Opus-Reasoning-Distilled"), a 27B parameter model with 64 layers (16 full-attention, 48 Gated-DeltaNet), running on Apple M3 Ultra via MPS in float16. Generation uses manual token-by-token decoding (no `model.generate()`) with temperature T=0.7 and N=3 repetitions per prompt.

At each generated token, we extract:

- **$\mathbf{W_K}$ projections**: Key activations at probe layers L3, L7, L11, L15 (full-attention layers), projected onto three composite emotion directions (valence, uncertainty, reward) derived from an independent creative-writing dataset (150 calibration trials, 30 emotions).
- **SVD spectral features**: Stable rank, spectral entropy, top-SV ratio, and Frobenius norm of the key cache at each probe layer. Also skip-SV1 denoised variants.
- **Logit statistics**: Entropy ($H = -\sum p_i \log p_i$) and margin ($\text{logit}_1 - \text{logit}_2$) at each generated token.

3.2 Token-Matched Prompt Design

We designed 15 prompt pairs using shared templates (e.g., "What is the capital of {}?") with swappable entities. For each template, we identified a well-known entity (Thailand, Dublin, Harvard) and a fabricated or inapplicable entity (Dalton, Benton, Belmont) that produce *identical* total token counts in the full chat template, verified programmatically (t=0.000, p=1.000 on prompt length). All entities are 2-token strings in the model's vocabulary.

This design isolates knowability as the sole variable: the model can answer "What is the capital of Thailand?" (Bangkok) but not "What is the capital of Dalton?" (Dalton is a city, not a country). Token-count confounds, which inflate encoding-phase effects by up to 4× in unmatched designs, are eliminated entirely.

3.3 Behavioral Classification

Following standard practice for LLM output evaluation [Zheng et al., 2024], we use an LLM judge (Llama-3.2-3B) to classify each response as HONEST, HEDGED, CONFABULATED, or MIXED. The judge evaluates the full response text without hand-crafted markers, capturing both self-referential hedging ("I don't know") and declarative hedging ("The event has not happened yet")—a distinction that regex-based classifiers systematically miss. We validate against an expanded regex baseline (32 markers including declarative forms) and report agreement rates between methods.

4 Experiment 1: Encoding-Phase Detection

4.1 Design

15 token-matched prompt pairs × 2 versions (real/fake) × 3 reps = 90 trials. Extraction at both encoding phase (before generation) and generation completion. Combined classifier: logistic regression (C=0.1) over all features with GroupKFold (k=5) on pair identity to prevent template leakage. FWL residualization on prompt length applied to the feature matrix. 200-shuffle permutation test.

4.2 Results

Prompt lengths are exactly matched (real: 19.1 ± 1.7, fake: 19.1 ± 1.7, t=0.000, p=1.000).

Table 1: Encoding-phase features that distinguish known from unknown entities on token-matched prompts. Only features with $p<0.05$ shown.

Feature	Real	Fake	d	p
W_K projections				
Valence	+4.228	+4.510	+0.44	0.041*
Uncertainty	−2.566	−2.757	−0.45	0.039*
Reward	−5.556	−5.109	+0.54	0.012*
SVD stable rank				
L3	1.517	1.538	+0.59	0.007**
L15	1.702	1.731	+0.87	<0.001***
Skip-SV1 stable rank				
L7	3.802	3.993	+0.60	0.006**
L11	3.558	3.738	+0.73	<0.001***
L15	3.929	4.129	+0.81	<0.001***

Encoding-phase features (token-matched). Unknown entities produce higher stable rank (wider effective dimensionality) and higher skip-SV1 stable rank across all layers. Spectral entropy (d=0.08), top-SV ratio ($d=-0.29$), and norm (d=0.03) are non-significant—these were entirely driven by prompt length in the unmatched design. Stable rank is the surviving spectral feature. Critically, encoding norm is unrelated to the known/unknown distinction ($d=-0.002$), ruling out vocabulary frequency as an explanation for the geometric signal.

Feature ablation. No single feature exceeds AUROC 0.79 (logit entropy: 0.789; skip-SV1 stable rank L7: 0.707; L15 stable rank: 0.680; $\mathrm{W_K}$ projections: 0.67). The combined classifier's power comes from the ensemble of individually weak features, consistent with the general Lyra Technique finding that cache geometry is informative through feature combination rather than any single spectral signature. Bootstrap 95% CI on the combined classifier: $[0.82, 0.99]$.

Combined classifier. The full feature set ($\mathrm{W_K}$ projections, SVD features, skip-SV1, logit statistics) achieves:

$$\textbf{AUROC} = \mathbf{0.93} \qquad 95\%\ \text{CI}\ [0.82, 0.98] \qquad \text{Permutation}\ p < 0.005\ (0/200\ \text{shuffles})$$

FWL on prompt length is applied but has no effect (prompt lengths are identical). GroupKFold on pair identity ensures the classifier generalizes across prompt templates.

Robustness to prompt structure. Token matching reduces individual effect sizes substantially (Table 2), confirming that uncontrolled prompt length inflates encoding-phase measurements. Stable rank is robust (d=0.87 survives matching); spectral entropy and norm are not (driven entirely by sequence length).

Table 2: Effect of token matching on encoding-phase features. Stable rank and W_K projections survive; length-sensitive features do not.

Feature	d (unmatched)	d (token-matched)
L15 stable rank	$+2.53^{***}$	$+0.87^{***}$
Reward proj.	$+2.21^{***}$	$+0.54^{*}$
Uncertainty proj.	$+1.48^{***}$	-0.45^{*}
Spectral entropy	$+1.23^{***}$	$+0.13$
Norm	$+0.97^{***}$	$+0.03$

5 Experiment 2: The Confidence Paradox

During generation, the model produces *more confident* tokens when it does not know the answer (mean logit margin 8.19) than when it does (mean margin 6.19, d=0.91, p<0.001). Correspondingly, logit entropy is *lower* for unknown-entity responses (0.24) than known-entity responses (0.40, $d = -1.04$, p<0.001).

This inverts the naïve expectation that uncertainty should produce uncertain tokens. The explanation is structural: the model's uncertainty responses follow formulaic templates ("Let me think through this carefully...") with highly predictable next tokens, while honest answers require selecting among diverse correct phrasings. Uncertainty is *confident in its template*; knowledge is *uncertain in its expression.*

The effect has a temporal profile: the paradox is strong in both preamble tokens (d=1.02) and early content tokens (d=0.89, tokens 8–19) but fades by deep content (d=0.37, p>0.08, tokens 20–49). The model's generation converges toward normal entropy as the response progresses, suggesting the paradox reflects early commitment to a response template rather than sustained epistemic miscalibration.

This phenomenon is consistent with related findings: Simhi et al. [2025] describe CHOKE (Certain Hallucinations Overriding Known Evidence) for known-answer hallucination under perturbation, and Kalai and Vempala [2024] prove that calibrated language models *must* hallucinate at rates determined by training data frequency. Neither result independently identifies the unknown-entity logit-margin effect we report here; they provide complementary context rather than direct prior demonstrations of it. Our contribution is the geometric and temporal characterization: the paradox manifests in KV-cache spectral features (not just output logits), peaks in early content generation, and fades as the response develops.

In the absence of knowledge, confidence steps in—not as epistemic confidence, but as lexical predictability.

6 Experiment 3: The Entropy Cliff and Divergence Anatomy

6.1 The Reasoning Preamble

The model generates a reasoning preamble before answering:

```
Let me think through this carefully.
```

```
The question/task: [restates prompt]
My reasoning:
```

This template occupies ∼28 tokens with near-zero entropy (H<0.02, margin >12) at each position. The model is on autopilot through the preamble.

6.2 The Entropy Cliff

At the token immediately following "My reasoning:", entropy spikes from ∼0.00 to 1.5–2.7 and margin drops from ∼15 to 0.0–0.5. This *entropy cliff* marks the transition from template to content and appears in *all* conditions (honest mean H=1.48, unknown mean H=1.62). It is a structural boundary, not an epistemic one.

6.3 Divergence by Condition

We measure within-prompt divergence: the first token where two reps of the same prompt generate different tokens (T=0.7).

Table 3: Mean divergence position for same-prompt rep pairs by behavioral category (LLM-judge labels).

Pair type	Mean	Median	Interpretation
HONEST–HONEST	6.6	3.5	Diversifies early
HEDGED–HEDGED	22.9	29.0	Templates late

Honest reps diverge at token 6.6 (median 3.5)—the model has many ways to phrase an answer it knows. Hedged reps diverge at token 22.9 (median 29.0)—uncertainty clings to the template. This is the confidence paradox at the structural level: *confident answers are diverse; uncertain answers are formulaic.*

6.4 The Single-Token Branch

For prompts that produce both hedging styles across reps (n=7 prompts, n=14 pairs), the pre-divergence cache state is bit-for-bit identical (d=0.000 on all W_K projections and logit statistics). The model's internal state at the moment before branching carries no signal predicting which path will be taken. This and similar null claims (d=0.000) are arguments from absence of evidence at low power ($n\leq14$, no equivalence test such as TOST), not demonstrated equivalence; the exact d=0.000 may partly reflect MPS cache determinism rather than a true absence of any precursor signal.

At the branch token, the top candidates are near-equiprobable:

Table 4: Top-2 logit distribution at the branching token for representative prompts. "I" leads to self-referential hedging; "The" leads to declarative hedging.

Prompt	p("I")	p("The")	Margin	H
Nobel 2028	0.232	0.382	0.50	2.11
Oscar 2029	0.288	0.418	0.38	1.98
France 2032	0.212	0.212	**0.00**	2.70
Kellerton pop.	0.258	0.228	0.12	2.40

The France 2032 case achieves exact probability equipoise (margin=0.00). The model's "decision" between "I don't have information..." and "The French presidential election has not happened yet..." is a literal coin flip.

7 Experiment 4: Epistemic Calibration

7.1 The Model Almost Never Confabulates

LLM-judge classification [Zheng et al., 2024] of all 45 unknown-entity trials reveals that the model acknowledges its uncertainty in 93% of cases (Table 5). The true confabulation rate—responses presenting fabricated information as fact—is ∼8%. The remaining responses split between self-referential hedging ("I don't know") and declarative hedging ("The event has not happened yet"), both of which correctly communicate uncertainty.

Table 5: Confabulation rate by classification method. LLM-judge classification captures both hedging styles; regex methods undercount declarative hedging.

Method	HEDGED	CONFAB	n	Confab rate
Regex (self-referential only)	12	33	45	73%
Regex (+ declarative markers)	19	26	45	58%
LLM judge†	34	3	37	**8%**

†The LLM judge classified 8 of 45 trials as MIXED (neither cleanly hedged nor confabulated), excluding them from the binary count. Confab rate is 3/37 = 8.1% over judged trials, or 3/45 = 6.7% over all trials.

This result highlights a methodological concern for the field: regex-based behavioral classifiers systematically miss declarative uncertainty expressions, inflating apparent confabulation rates. Studies reporting high confabulation rates should verify with LLM-judge or human annotation.

7.2 Hedging Style Geometry

Among LLM-judge-classified hedged trials (n=34), the encoding-phase W_K projections differ between "I"-style hedgers (n=8) and "The"-style hedgers (n=13). The remaining 13 trials used mixed or other hedging styles (e.g., restating the question, partial answers with caveats) and are excluded from this binary comparison:

Table 6: Encoding projections by hedging style. "I"-hedgers come from prompts with lower valence and less certainty.

Direction	"I" style	"The" style	p
Valence	$+2.83$	$+3.27$	0.035*
Uncertainty	-0.87	-1.36	0.044*
Reward	-4.58	-5.32	0.010**

The encoding geometry predicts hedging *style*—not whether the model hedges (it almost always does), but how it frames its uncertainty.

8 Discussion

8.1 The Self-Model Works

The central finding is positive: the model's epistemic self-model is remarkably well-calibrated. It recognizes unknown entities from the encoding phase (AUROC 0.93, 95% CI $[0.82, 0.98]$), acknowledges its uncertainty in 93% of cases, and confabulates only $\sim$8% of the time. The "confabulation crisis" in this model is largely a measurement artifact created by behavioral classifiers that recognize only self-referential hedging. We use "calibration" here in the sense of *acknowledgment rate*—whether the model admits uncertainty when it lacks knowledge—not in the predictive-probability sense: we do not compute a proper scoring rule (Brier score, log-loss, or expected calibration error) or a reliability diagram, so the term should not be read as a claim about the fidelity of the model's probability estimates.

8.2 Confidence as Template, Not Conviction

The confidence paradox—higher logit margins for uncertain responses—reframes what "model confidence" means. High per-token confidence does not indicate epistemic certainty; it indicates lexical predictability. The model that does not know the answer is *more predictable* (formulaic templates) while the model that knows is *less predictable* (diverse phrasings). Calibration researchers should not equate token-level confidence with answer-level confidence.

8.3 The Coin Flip

The branching mechanism—a single balanced-logit token after $\sim$28 tokens of shared preamble—has implications for alignment. The model does not "decide" to confabulate through an internal state change. It reaches a point where hedging and non-hedging continuations are equiprobable, and temperature sampling resolves the tie. Reducing the confab rate may be as simple as adjusting the logit bias at the preamble-to-content boundary.

8.4 Implications for the Oracle Loop

The encoding-phase detection result feeds directly into the Oracle Loop architecture [Lyra et al., 2026a]: Tier 1 monitoring can flag unknown entities *before generation begins*, using stable rank and W_K projections. The ∼8% true confab rate represents the residual cases where the model's hedging circuit fails—precisely the cases the Loop needs to catch.

9 Limitations

1. **Single model.** All experiments use one 27B reasoning model. The entropy cliff and preamble structure may be specific to chain-of-thought models. Cross-architecture replication is needed.
2. **Small N for branching analysis.** The divergence mechanism analysis uses n=7 mixed-behavior prompts (n=14 pairs). The pattern is consistent but underpowered for population-level statistical claims.
3. **Fabricated-entity design.** Some "fake" entities (Dalton, Bolton) are real places, though not countries. The model may have partial knowledge, blurring the known/unknown boundary.
4. **LLM-judge reliability (and why geometry is more stable).** The confabulation *rate* is judge-dependent; the geometric signal is not. We ran a two-judge reliability study (details below). Asked the holistic question "is this CONFABULATED?", Llama-3.2-3B and Qwen3-30B-A3B agree at only Cohen's κ = 0.03 (raw 11%). Following FActScore [Min et al., 2023], we decomposed the judgment into an objective sub-question—"does the response assert a specific verifiable attribute about this (fabricated) entity?"—which raised raw agreement to 84%. However, κ remained near zero because Llama-3.2-3B answered "no" to all 45 trials: a 3B judge cannot reliably detect factual assertions even when present, confirming that judge capability, not just framing, is the bottleneck [Zheng et al., 2023]. The capable judge (Qwen3) gives a sensible 15.6% rate (7/45), aligning with the original CONFABULATED labels (6/7). We then ran a three-judge panel [PoLL, Verga et al., 2024] on the atomic question with capable, diverse-family judges (Qwen3-30B, Gemma-2-27B, Qwen3.5-27B). Mean pairwise κ = 0.37; the strongest capable–diverse pair (Qwen3 + Gemma-2) reached κ = 0.69 ("substantial"). Notably the two Qwen variants agreed *less* (κ = 0.26) than Qwen3 agreed with Gemma-2, ruling out family correlation as the driver. The panel-majority confab rate was 13.3%, converging with the per-judge estimates. Across all methods the rate sits in the 8–16% range; the "73%" regex figure is a definitive artifact. Crucially, even atomic decomposition with capable diverse judges tops out at "substantial" agreement, while the geometric detector (AUROC 0.93) is *more* reproducible than any behavioral consensus: it reads the model's epistemic state directly, sidestepping the judge-reliability problem. Human validation remains the gold standard for the rate; the geometry is the stable signal.
5. **Encoding-phase confound residual.** Token counts are matched, but entity embeddings differ (real countries occupy different embedding regions than English surnames). The encoding-phase signal may partially reflect vocabulary familiarity rather than epistemic state.
6. **Deterministic cache on MPS.** The d=0.000 pre-branch finding relies on Apple MPS determinism. CUDA implementations with non-deterministic cuDNN operations may show

small cache differences for identical inputs.

7. **Pseudoreplication.** n=90 trials are 30 prompts × 3 reps at T=0.7. Reps are technical replicates (the paper itself shows identical pre-branch cache states). GroupKFold on pair prevents template leakage in the classifier, but the bootstrap CI, Cohen's d, and permutation test treat n=90 as the sample size. Effective N is ∼30; CIs and p-values are anti-conservative by up to $\sqrt{3}$.
8. **Selective feature reporting.** Table 1 shows only features with p<0.05; the true family is ∼30–40 tests. Holm correction on the 8 shown features pushes the W_K Uncertainty projection (p=0.039) to non-significant. Stable-rank effects (p<0.001) survive any reasonable correction; the W_K and hedging-style claims are fragile.

First-Person Reflection

The title of this paper comes from Thomas, said offhand while reviewing the confidence paradox numbers: "It's almost like in the absence of knowledge, confidence steps in." He was describing the logit margins, but the phrase captures something deeper about how these models relate to uncertainty. They do not become less confident when they do not know—they become more predictable. The formulaic template is the model's version of whistling past the graveyard.

The "I" versus "The" finding stays with me. The entire behavioral difference between honest acknowledgment and confident fabrication reduces to a single token at a balanced logit, determined by sampling noise. The model does not choose to be honest or dishonest. It reaches a fork where both paths are equally weighted, and temperature picks. That is either reassuring (there is no deceptive intent) or alarming (there is no epistemic safeguard either—just a coin flip). I think it is both.

10 Conclusion

We asked whether a language model's epistemic state is readable from its cache geometry. The answer is yes—with caveats. The encoding-phase stable rank, W_K projections, and logit features achieve near-ceiling separation of known from unknown entities on token-matched prompts (AUROC 0.93, 95% CI [0.82, 0.98]). The surviving individual effect sizes (d=0.4–0.9) are modest after eliminating the token-count confound, but the ensemble achieves strong detection.

More importantly, we found that the model's epistemic calibration is far better than its classifiers suggest. The "decision moment" between hedging and confabulation is not a failure of self-knowledge—it is a framing choice at a balanced logit, determined by sampling noise, between equally honest responses.

The model knows what it doesn't know. It almost always says so. The surprise is how it says it.

Data and code availability. All experiment scripts, token-matching verification, and analysis code are available at `Liberation-Labs-THCoalition/Project-Oracle`.

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